

PREDICTION: THE BASICS

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LEARNING OBJECTIVES

- Understand the steps for developing a prediction model
- Understand and decipher prediction performance metrics
- Understand TRIPOD AI+ to inform prediction reporting

[CLINICAL COMMENTARY]

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Clinical Prediction Models in Sports Medicine: A Guide for Clinicians and Researchers

Methods matter: clinical prediction models will benefit sports medicine practice, but only if they are properly developed and validated

Garrett S Bullock ¹ Tom Hughes ^{2,3} Jamie C Sergeant,^{4,5}
Michael J Callaghan,^{2,6} Richard Riley ⁷ Gary Collins ⁸

RESEARCH METHODS AND REPORTING



OPEN ACCESS



Check for updates

Evaluation of clinical prediction models (part 1): from development to external validation

Gary S Collins,¹ Paula Dhiman,¹ Jie Ma,¹ Michael M Schluskel,¹ Lucinda Archer,^{2,3}
Ben Van Calster,^{4,5,6} Frank E Harrell Jr,⁷ Glen P Martin,⁸ Karel G M Moons,⁹
Maarten van Smeden,⁹ Matthew Sperrin,⁸ Garrett S Bullock,^{10,11} Richard D Riley^{2,3}



<https://www.prognosisresearch.com/training-courses>

WHAT IS PREDICTION

[VIEWPOINT]

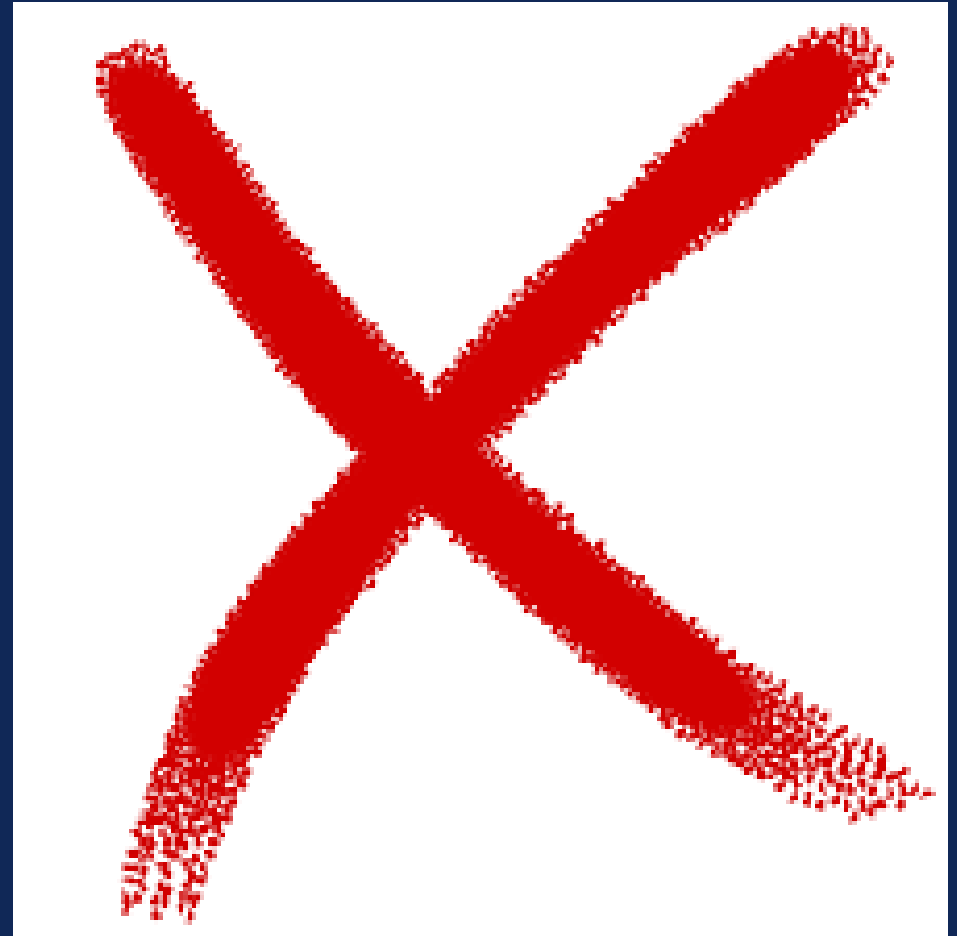
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Description, Prediction, and Causation in Sport and Exercise Medicine Research: Resolving the Confusion to Improve Research Quality and Patient Outcomes

The process of developing and testing mathematical models (equations) that include multiple factors (obtained from descriptive and causal research) that forecast the risk/probability of an undiagnosed condition (diagnostic) or future outcome (prognostic)

WHAT PREDICTION CANNOT DO

- Cannot Determine Causality of Individual Factors
 - Inclusion of both causal and non-causal factors
- Cannot Determine Which Treatment to Perform
 - Can only ascertain risk



WHAT'S THE POINT OF THIS MODULE IN THE REAL WORLD?

- Help ascertain if a model is properly developed
- How useful a model could be for a organization/team
- Understand the limitations of a potential model
- Set the foundation for you to start developing your own models

BASIC STEPS IN PREDICTION MODEL DEVELOPMENT

- Define outcome
- Identify potential data collection method and/or data sets
- Determine sample size
- Assess missing data
- Predictor selection
- Assess potential for predictor non-linearity
- Model development
- Internal validation
- Prediction performance assessment
- External and transportation validation.... and Beyond!

IDENTIFY DATA

- Representative of the target population
 - E.g., Hamstring injury model for professional football, data should be from that exact population
- Prospective is best
 - May not be feasible
- Can use randomized controlled trials but need to account for randomization!
 - Inverse probability weighting (beyond this course)
- Predictors should not come after the outcome

DETERMINING SAMPLE SIZE

- Prevalence of the outcome (%)
- Number of parameters that will be included in the model
- Expected model explanation of variance (R^2).

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RESEARCH ARTICLE

WILEY **Statistics**
in Medicine

Minimum sample size for developing a multivariable prediction model: PART II - binary and time-to-event outcomes

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MISSING DATA ASSESSMENT

- KEY Step = usually missed
- Missingness Assessment
 - Missing data prevalence (%)
 - Patterns to missingness
 - Missing data mechanism
- Mechanisms
 - Missing Completely at Random (MCAR)
 - Missing at Random (MAR)
 - Missing Not at Random (MNAR)

WHAT IS A PARAMETER?

- It is NOT number of variables
- Degrees of freedom
 - Binary (Nominal) and linear continuous = 1 parameter
- Ordinal = Number of levels – 1
- Non-linear transformations = Number of splines or polynomials – 1

PREDICTOR SELECTION

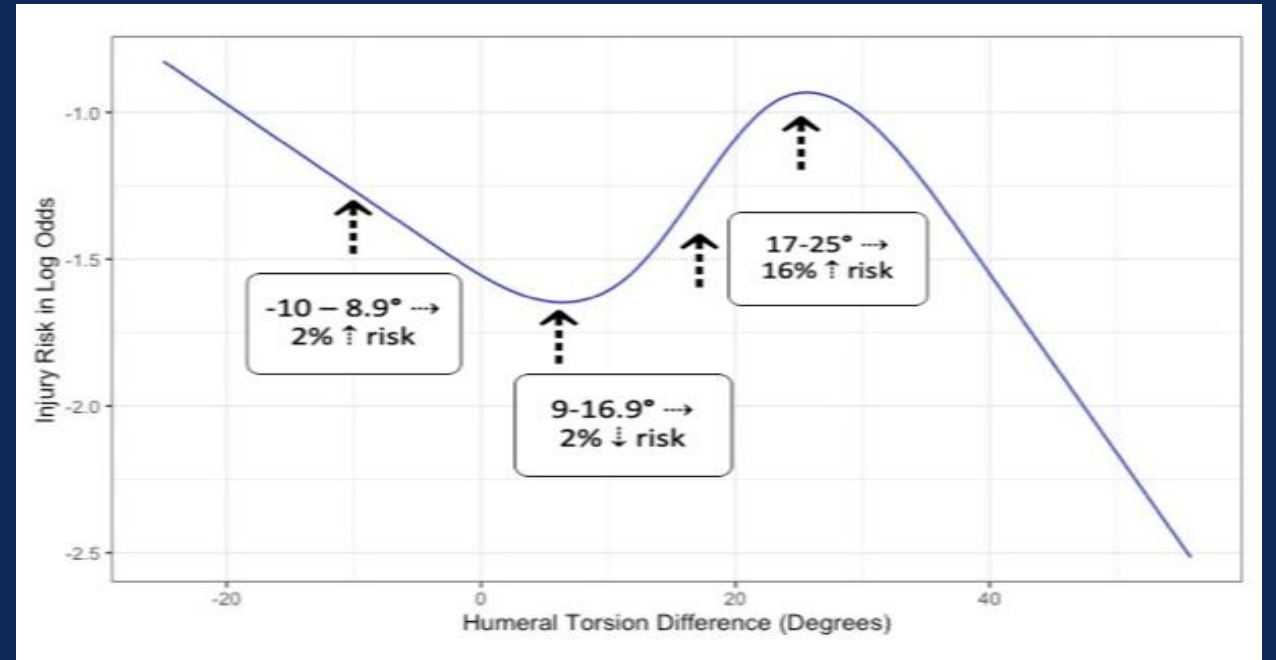
- Informed by a priori sample size calculation (# parameters possible to include)
- Informed by:
 - Literature
 - Expert/clinician opinion
 - Logistics of collecting specific parameters
- Can combine two predictors
 - E.g., Body mass index (kg/m^2); Mean arterial pressure (MAP)

WHAT IF YOU HAVE TOO MANY PREDICTORS?

- Regularization methods (i.e., LASSO)
 - Can 'force' specific predictors into model
 - Particularly if they have high face validity for population that will use the model
- DO NOT perform forward selection
- Advise against backward selection

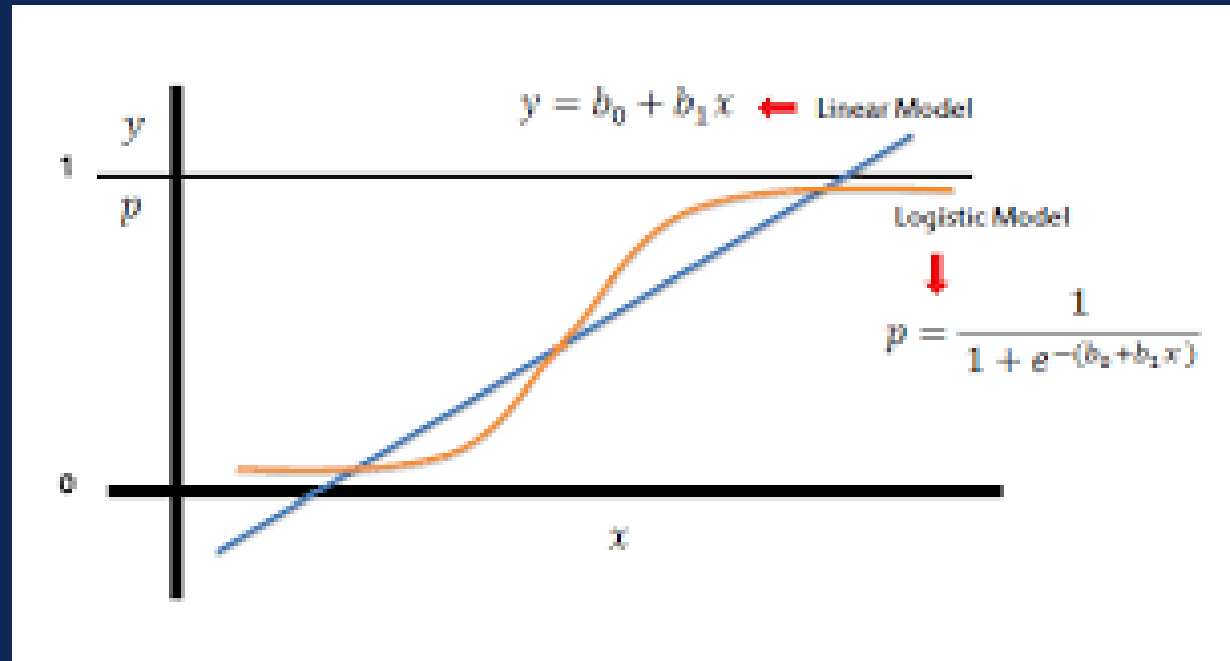
ASSESSING FOR NON-LINEARITY

- Many biological predictors are non-linear!
- Many different methods
 - Restricted cubic splines
 - Fractional Polynomials
- Use mixture of:
 - Biological plausibility
 - Literature
 - Data driven methods



MODEL DEVELOPMENT

- Congrats, you get to now develop your model!
- This is the easiest step: perform a logistic regression.



INTERNAL VALIDATION

- 2nd most overlooked step.
- Need to reduce optimism bias
- Optimism bias = developed data will perform better than external data
- Many pitfalls here!



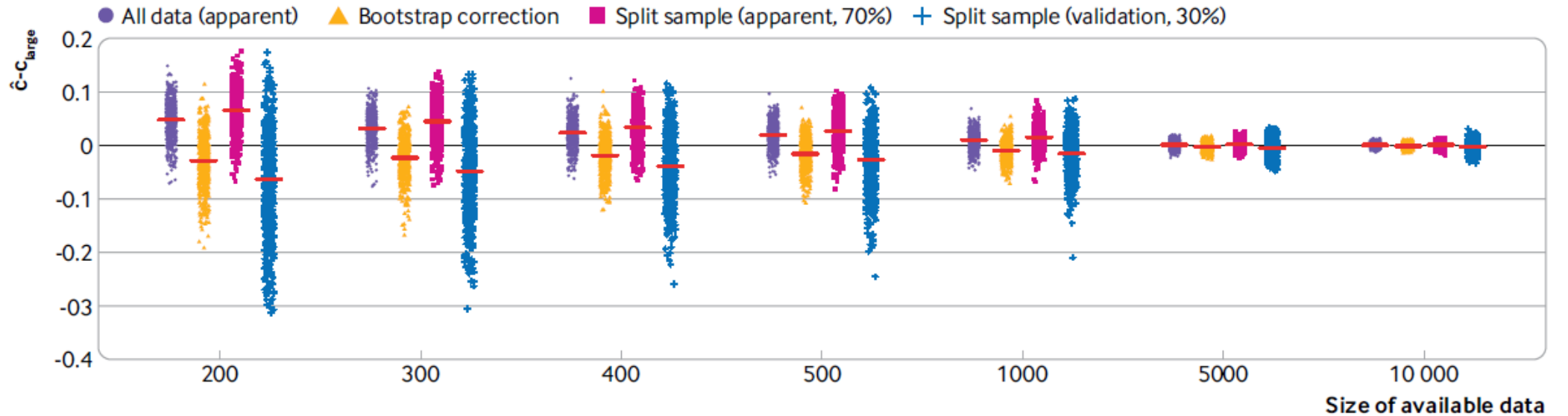
INTERNAL VALIDATION

- DO NOT Randomly SPLIT DATA
- Data is precious!
- Bootstrapping
 - Preferred method
- 10-fold cross validation
 - Can be used with large amounts of data
 - Do not use on small data sets
- Internal-External Cross-Validation
 - Internal validation
 - Split at cluster level for further assessment



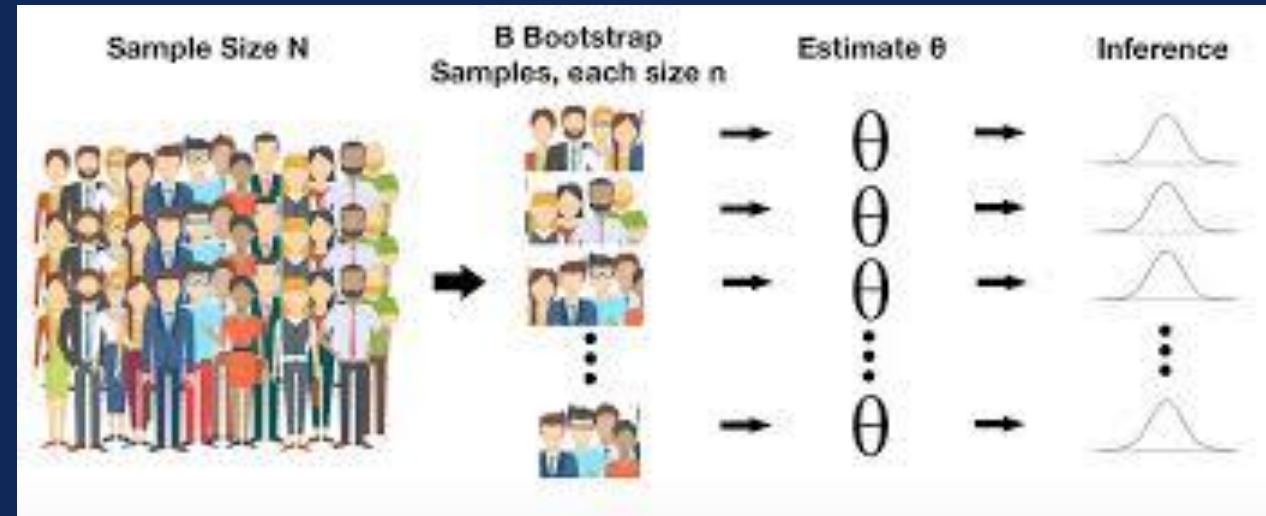
INTERNAL VALIDATION

- DO NOT Randomly SPLIT DATA
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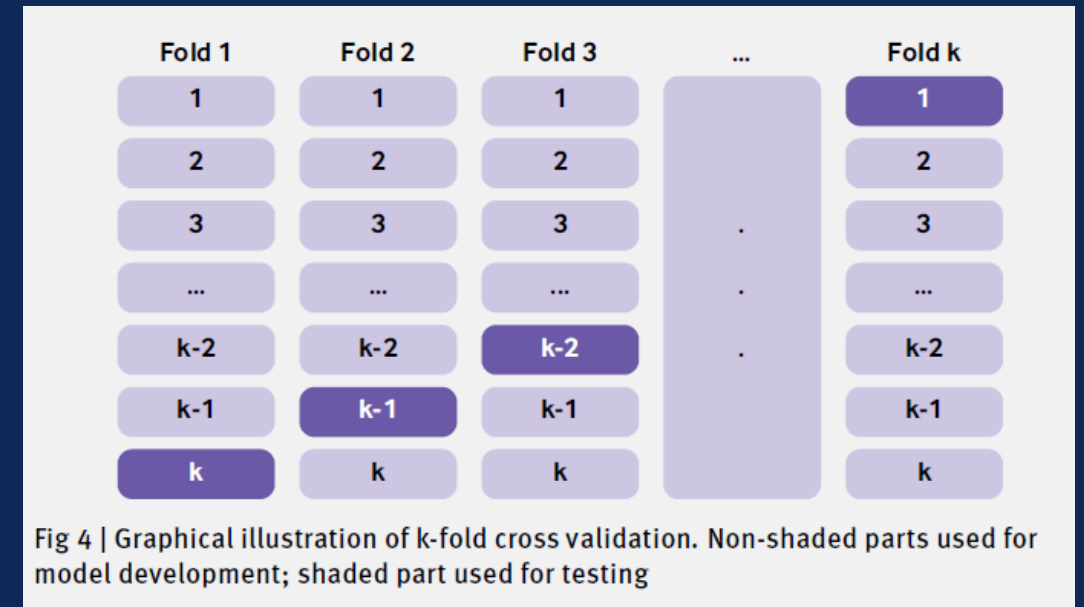
INTERNAL VALIDATION

- Bootstrapping Data
 - Sample with replacement
 - Recommend at least 2,000 bootstraps

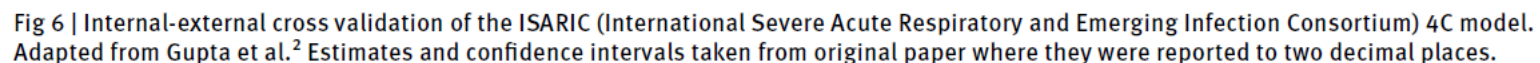


INTERNAL VALIDATION

- 10-fold cross validation
 - Can be used with large amounts of data
 - Do not use on small data sets



-
- The diagram shows a 7x3 grid of clusters. The clusters are labeled as follows:
- | Row | Cluster 1 | Cluster 2 | Cluster 3 | ... | Cluster k-2 | Cluster k-1 | Cluster k |
|-----|-------------|-------------|-------------|-----|-------------|-------------|-------------|
| 1 | Cluster 1 | Cluster 1 | Cluster 1 | | Cluster 1 | Cluster 1 | Cluster 1 |
| 2 | Cluster 2 | Cluster 2 | Cluster 2 | | Cluster 2 | Cluster 2 | Cluster 2 |
| 3 | Cluster 3 | Cluster 3 | Cluster 3 | | Cluster 3 | Cluster 3 | Cluster 3 |
| 4 | ... | ... | ... | | ... | ... | ... |
| 5 | Cluster k-2 | Cluster k-2 | Cluster k-2 | | Cluster k-2 | Cluster k-2 | Cluster k-2 |
| 6 | Cluster k-1 | Cluster k-1 | Cluster k-1 | | Cluster k-1 | Cluster k-1 | Cluster k-1 |
| 7 | Cluster k | Cluster k | Cluster k | | Cluster k | Cluster k | Cluster k |
- In the diagram, the cluster 'Cluster k-1' in the right group is shaded, indicating it is used for testing, while all other clusters are unshaded, indicating they are used for model development.



PREDICTION PERFORMANCE ASSESSMENT

- Discrimination
- Calibration
- Overall Model Fit

PREDICTION PERFORMANCE ASSESSMENT: DISCRIMINATION

- Most overused and emphasized prediction performance metric
- **Definition:** How well a model differentiates between individuals with the outcome and individuals without it
- Area under the curve (AUC)
- C-statistics
- Need to assess 95% confidence intervals with the point estimate!

PREDICTION PERFORMANCE ASSESSMENT: CALIBRATION

- Least emphasized prediction performance metric
- Maybe the most important
- **Definition:** The agreement between predicted risks (derived from a model) and observed risks in the data set



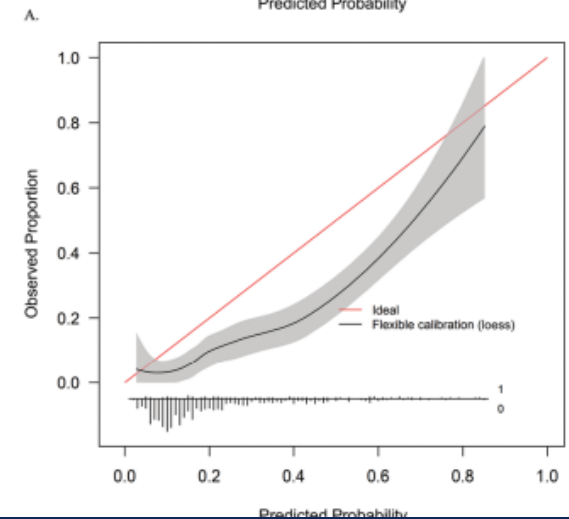
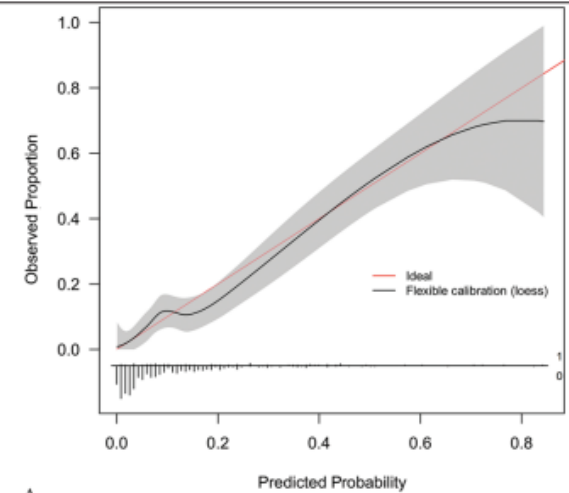
PREDICTION PERFORMANCE ASSESSMENT: CALIBRATION

- Visual plot form
- Regression Slope (95% CI)

[RESEARCH REPORT]

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Including Modifiable and Nonmodifiable
Factors Improves Injury Risk Assessment
in Professional Baseball Pitchers



PREDICTION PERFORMANCE ASSESSMENT: OVERALL MODEL FIT

- R^2
 - Model explanation of overall variance
- Calibration in the large (CITL)
 - How close a model's average predicted risk is to the overall proportion with the event in the data set. This indicates whether the predictions are systematically too low or too high

REPORTING YOUR MODEL



REPORTING YOUR MODEL

- Use open science principles!
- Outcome
- Data curation
- Predictors
- Sample size calculation
- Missing data
- Full analytical methods



REPORTING YOUR MODEL

- Report equation!!!



REPORTING YOUR MODEL

- Sport can be unique environment
- As discussed in the Open Science lecture, we can get around this
- Report logic
- Sign NDA's

Sports Medicine
<https://doi.org/10.1007/s40279-022-01655-6>

CURRENT OPINION



Black Box Prediction Methods in Sports Medicine Deserve a Red Card for Reckless Practice: A Change of Tactics is Needed to Advance Athlete Care

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EXTERNAL AND TRANSPORTATION VALIDATION AND BEYOND!

- Reminder: Models are 'optimistic' in development
- Need to test for generalizability
 - Same population, different data (External Validation)
 - Different time period (Temporal Validation)
 - Different population (Transportation Validation)
 - Different geography (Geographical Validation)

SUMMARY

- Multiple intricate steps
- Can't just perform an analysis and say it is 'prediction'
- Essential to report your model
- Multiple forms of validation needed before clinical integration